An elementary proof of Bézout's theorem on algebraic curves

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Introduction

This proof is inspired by a book named "Rational points on elliptic curves" by Tate and Silverman. I tried to make it rather elementary and to give some theory to introduce the different objects.

Let's state the theorem now :

Theorem. Let $\mathcal{C}_1, \mathcal{C}_2$ be two algebraic projective curves defined over an algebraically closed field, without common component, of respective degree m and n, then $\#\mathcal{C}_1 \cap \mathcal{C}_2 = mn$ when multiplicity is taken into account.

Clearly, we lack some definition of "projective curve", or "multiplicity", but they will be given through the text.

1 Some projective geometry

There are two classical ways to introduce the projective plane: an algebraic one and a geometric one.

For the algebraic approach, consider the Fermat equation $x^n + y^n = z^n$. Looking for integer solutions is equivalent to looking for rational ones of $x^n + y^n = 1$. A key observation is that if (x, y, z) is a solution, then so is (mx, my, mz) for any non-zero integer m. These triples should therefore represent the same point: only their direction in k^3 matters. This motivates the definition of the projective plane $\mathbb{P}^2$ over a field k as

$$\mathbb{P}^2 = (k^3 \setminus \{0\})/\sim,$$

where $(X : Y : Z) \sim (\lambda X : \lambda Y : \lambda Z)$ for all $\lambda \in k^\times$. The reader may define $\mathbb{P}^n$ similarly.

For the geometric viewpoint, we want a space in which Bézout's theorem holds for lines. In the affine plane k^2, the only obstruction comes from parallel lines, which never meet. Projective geometry fixes this by adding a point at infinity for each possible direction. This is the familiar phenomenon of the two sides of a straight road appearing to meet at a vanishing point. Thus, geometrically,

$$\mathbb{P}^2 = k^2 \cup \{\text{directions of lines in } k^2\}.$$

To make this precise, consider the affine chart $Z = 1$. Any point $[X : Y : Z]$ with $Z \neq 0$ can be uniquely written as

$$[X : Y : Z] = \big[X/Z : Y/Z : 1\big],$$

which identifies the open subset $\{Z \neq 0\}$ in $\mathbb{P}^2$ with the affine plane k^2 via

$$(x, y) \longleftrightarrow [x : y : 1].$$

The complement of this chart is the line $\{Z = 0\}$, called the *line at infinity*. It consists exactly of the directions of affine lines: each point $[X : Y : 0]$ corresponds to a slope or direction in k^2. Parallel affine lines meet at the point of $\{Z = 0\}$ representing their common direction.

We therefore obtain the geometric decomposition

$$\mathbb{P}^2 = \{Z \neq 0\} \cup \{Z = 0\} = k^2 \cup \mathbb{P}^1,$$

where $\mathbb{P}^1$ parametrizes the directions of lines in the affine plane.

To make this clear, here is a recap array :

Algebraic definition		Geometric definition
$k^3 \setminus \{0\}/\sim$	$\leftrightarrow$	$k^2 \cup \mathbb{P}^1$
(a, b, c)	$\rightarrow$	$\begin{cases} (a/c, b/c) \in k^2 & \text{if } c \neq 0 \\ (a, b) \in \mathbb{P}^1 & \text{if } c = 0 \end{cases}$
$(x, y, 1)$	$\leftarrow$	$(x, y) \in k^2$
$(x, y, 0)$	$\leftarrow$	$(x, y) \in \mathbb{P}^1$

We now introduce the first fundamental notion: a **projective** (algebraic) curve. Let $f \in k[X, Y]$ be a polynomial of degree d, which we may write as

$$f(X, Y) = \sum_{i+j \leq d} a_{i,j} X^i Y^j.$$

Our goal is to extend the affine curve $\mathcal{C} = \{f = 0\}$ to a curve in $\mathbb{P}^2$. To do so, we homogenize f by defining

$$F(X, Y, Z) = \sum_{i+j \leq d} a_{i,j} X^i Y^j Z^{d-i-j}.$$

Then F is homogeneous of degree d, and satisfies

$$F(X, Y, 1) = f(X, Y).$$

For a point $(x : y : z) \in \mathbb{P}^2$, the equation $F(x, y, z) = 0$ is well defined: indeed, since $(x : y : z) \sim (tx : ty : tz)$ for any $t \in k^\times$, we must have

$$F(tx, ty, tz) = 0 \quad \text{whenever} \quad F(x, y, z) = 0.$$

This holds precisely because F is homogeneous:

$$F(tx, ty, tz) = t^d F(x, y, z).$$

Thus $F = 0$ defines a projective curve in $\mathbb{P}^2$ which extends the affine curve $\mathcal{C}$. More precisely,

$$(x, y) \in \mathcal{C} \quad \Longleftrightarrow \quad (x : y : 1) \in \{(a : b : c) \in \mathbb{P}^2 \mid F(a, b, c) = 0\}.$$

The projective curve also contains additional points at infinity, namely those of the form $(x : y : 0)$ satisfying $F(x, y, 0) = 0$.

Throughout the rest of this paper, we assume that k is an algebraically closed field (in particular, k is infinite).

2 The proof

The proof can be divided into five steps, but we first introduce the notation. Let $f_1, f_2 \in R := k[X, Y]$ be two polynomials of respective degrees n_1 and n_2, assumed to have no common component (i.e. any $g \in R$ dividing both f_1 and f_2 must lie in k). Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be the corresponding projective curves defined in the previous paragraph.

1. We first prove the inequalities

   $$\#(\mathcal{C}_1 \cap \mathcal{C}_2 \cap k^2) \overset{(A)}{\leq} \dim\big(R/(f_1, f_2)\big) \overset{(B)}{\leq} n_1 n_2,$$

   which already shows that the number of affine intersection points of $\mathcal{C}_1 \cap \mathcal{C}_2$ is at most $n_1 n_2$.

2. We then show that (B) is in fact an equality when $\mathcal{C}_1$ and $\mathcal{C}_2$ do not meet at infinity.

3. We introduce a suitable definition of intersection multiplicity and verify that it agrees with geometric intuition on a quite general example.

4. We prove that
$$\sum_{P \in \mathcal{C}_1 \cap \mathcal{C}_2 \cap k^2} I(\mathcal{C}_1 \cap \mathcal{C}_2, P) \;=\; \dim\big(R/(f_1, f_2)\big).$$

5. Finally, we show that the definition of intersection multiplicity is invariant under projective changes of coordinates. Using this, we choose a projective transformation sending a line disjoint from $\mathcal{C}_1$ and $\mathcal{C}_2$ to the line at infinity, which allows us to conclude.

The assumption that k is algebraically closed is only used in steps (2) and (4).

2.1 Step 1

We begin with inequality (A), using the fact that if $(x_1, \dots, x_n)$ is a linearly independent family in a vector space E, then $\dim(E) \geq n$. Let $P_1, \dots, P_m$ be points of $\mathcal{C}_1 \cap \mathcal{C}_2 \cap k^2$. We construct polynomials $(h_i)_{i \leq m}$ in R which are linearly independent modulo (f_1, f_2). For each $i \in [\![1, m]\!]$ and each $j \neq i$, choose a line
$$L_j = \{(x, y, z) \in \mathbb{P}^2 \mid g_j(x, y, z) = a_j x + b_j y + c_j z = 0\}$$
passing through P_j but not through P_i. Define
$$h_i = \prod_{j \neq i} \frac{g_j}{g_j(P_i)}.$$
This is a projective analogue of Lagrange interpolation: we have
$$h_i(P_i) = 1 \qquad \text{and} \qquad h_i(P_j) = 0 \text{ for } j \neq i.$$

To prove independence modulo (f_1, f_2), let $(\lambda_i)_{i \leq m} \in k^m$ and $g_1, g_2 \in R$ satisfy
$$\sum_{i \leq m} \lambda_i h_i = g_1 f_1 + g_2 f_2.$$
Evaluating at each P_i gives $\lambda_i = 0$ for all i, hence the h_i are linearly independent modulo (f_1, f_2). Therefore,
$$\boxed{\#(\mathcal{C}_1 \cap \mathcal{C}_2 \cap k^2) \leq \dim\big(R/(f_1, f_2)\big)}.$$

We now introduce some notations. Let R_d be the k-vector space of polynomials of degree $\leq d$, and define
$$W_d := R_{d-n_1} f_1 + R_{d-n_2} f_2.$$

Note that $W_d = 0$ if $d < \max(n_1, n_2)$, and in all cases $W_d \subset (f_1, f_2)$. Let

$$\phi(d) = \frac{1}{2}(d+1)(d+2),$$

so that $\dim(R_d) = \phi(d)$.

Assume $d \geq n_1 + n_2$. We claim that

$$R_{d-n_1} f_1 \cap R_{d-n_2} f_2 = R_{d-n_1-n_2} f_1 f_2.$$

The inclusion $\supset$ is clear. For the reverse inclusion, let $g \in R_{d-n_1} f_1 \cap R_{d-n_2} f_2$. Then $g = h_1 f_1 = h_2 f_2$ for some $h_1 \in R_{d-n_1}$ and $h_2 \in R_{d-n_2}$. Since f_1 and f_2 have no common component, f_1 divides h_2, so $g = h_3 f_1 f_2$ for some $h_3 \in R_{d-n_1-n_2}$.

We now compute:

$$\begin{aligned}
\dim(R_d) - \dim(W_d) &= \phi(d) - (\dim(R_{d-n_1} f_1) + \dim(R_{d-n_2} f_2) - \dim(R_{d-n_1} f_1 \cap R_{d-n_2} f_2)) \\
&= \phi(d) - (\dim(R_{d-n_1}) + \dim(R_{d-n_2}) - \dim(R_{d-n_1-n_2} f_1 f_2)) \\
&= \phi(d) - (\phi(d-n_1) + \phi(d-n_2) - \phi(d-n_1-n_2)) \\
&= n_1 n_2
\end{aligned}$$

We can now prove inequality (B). Let $(g_i)_{i \leq n_1 n_2 + 1} \subset R$. Choose $d \in \mathbb{N}$ large enough so that all g_i lie in R_d and $d \geq n_1 + n_2$. From the previous computation, we have $\dim(R_d/W_d) = n_1 n_2$, hence

$$\mathrm{Vect}(g_i) \cap W_d \neq \{0\}.$$

Thus there exists a non-zero family (λ_i) such that

$$\sum \lambda_i g_i \in W_d \subset (f_1, f_2),$$

showing that the g_i are linearly dependent modulo (f_1, f_2). Therefore,

$$\boxed{\dim\big(R/(f_1, f_2)\big) \leq n_1 n_2}.$$

2.2 Step 2

In this step, we assume that $\mathcal{C}_1$ and $\mathcal{C}_2$ do not meet at infinity.

To use this hypothesis, we first describe the zeros at infinity of a polynomial $f = \sum_{i,j} c_{i,j} X^i Y^j \in R$ of degree n. We denote by f^* its homogeneous part of highest degree:

$$f^* = \sum_{i+j=n} c_{i,j} X^i Y^j = Y^n \sum_{i+j=n} c_{i,j} \left(\frac{X}{Y}\right)^i.$$

Since k is algebraically closed, f^* factors into linear forms (viewing it as a polynomial in X/Y):

$$f^* = \prod_{i=1}^{n} (a_i X + b_i Y).$$

When we homogenize f, the highest-degree part f^* remains unchanged. Points at infinity are of the form $(x, y, 0)$ with $(x, y) \neq (0, 0)$, and for such points we have

$$F(x, y, 0) = f^*(x, y).$$

Thus

$$\begin{aligned} F(x,y,0) = 0 &\iff f^*(x,y) = 0 \\ &\iff \exists i \in [\![1, n]\!],\ a_i x + b_i y = 0 \\ &\iff \exists i \in [\![1, n]\!],\ (x, y, 0) = (b_i, -a_i, 0). \end{aligned}$$

We now show that f_1^* **and** f_2^* **are coprime**. If they were not, then since both factor into linear forms, they would share at least one factor $aX + bY$ with $(a, b) \neq (0, 0)$. By the previous remark, this would imply $(b, -a, 0) \in \mathcal{C}_1 \cap \mathcal{C}_2 \cap \mathbb{P}^1$, contradicting our assumption that the curves do not meet at infinity.

We now prove that $(f_1, f_2) \cap R_d = W_d$ for $d \geq n_1 + n_2$. The inclusion $W_d \subset (f_1, f_2) \cap R_d$ is clear. For the reverse inclusion, let

$$f = g_1 f_1 + g_2 f_2 \in (f_1, f_2) \cap R_d,$$

where g_1 and g_2 are chosen with minimal possible degrees. Assume $\deg(g_1) > d - n_1$. Looking at the highest-degree terms gives

$$g_1^* f_1^* + g_2^* f_2^* = 0, \text{ i.e } g_1^* f_1^* = -g_2^* f_2^*.$$

Since f_1^* and f_2^* are coprime, there exists $h \in R$ such that

$$g_1^* = h f_2^* \qquad \text{and} \qquad g_2^* = -h f_1^*.$$

Because * extracts the highest-degree part, we deduce

$$\deg(g_1 - h f_2) < \deg(g_1) \quad \text{and} \quad \deg(g_2 + h f_1) < \deg(g_2).$$

But

$$(g_1 - h f_2) f_1 + (g_2 + h f_1) f_2 = g_1 f_1 + g_2 f_2 = f,$$

contradicting the minimality of the degrees of g_1 and g_2. Thus $g_i \in R_{d-n_i}$ and $f \in W_d$.

We now consider the map

$$\begin{array}{rccc} \phi : & R_d/W_d & \longrightarrow & R/(f_1, f_2) \\ & f \bmod W_d & \longmapsto & f \bmod (f_1, f_2) \end{array}.$$

If $f \in R_d$ satisfies $f \equiv 0 \bmod (f_1, f_2)$, then $f \in (f_1, f_2) \cap R_d = W_d$, hence $f \equiv 0 \bmod W_d$. Thus ϕ is injective. Since $\dim(R_d/W_d) = n_1 n_2$, we obtain

$$n_1 n_2 = \dim(R_d/W_d) \leq \dim(R/(f_1, f_2)),$$

which shows that (B) is an equality:

$$\boxed{\dim(R/(f_1, f_2)) = n_1 n_2}.$$

2.3 Step 3

We now introduce a suitable definition of intersection multiplicity. The guiding intuition is that the ring of functions defined near a point encodes most of the local geometric information.

Let $K = k(X, Y)$ be the field of rational functions in two variables. For any point $P \in k^2$, we define

$$\mathcal{O}_P := \{f/g \in K \mid f, g \in R,\ g(P) \neq 0\}.$$

This is exactly the localization of $R = k[X, Y]$ at the maximal ideal

$$M := \{f \in R \mid f(P) = 0\},$$

so that $\mathcal{O}_P = R_M$. The evaluation map $f/g \mapsto f(P)/g(P)$ is well defined on $\mathcal{O}_P$, and its kernel is the unique maximal ideal

$$M_P := M\mathcal{O}_P = \{f/g \in \mathcal{O}_P \mid f(P) = 0\}.$$

Thus $\mathcal{O}_P$ is a local ring with maximal ideal M_P, and we have $\mathcal{O}_P = k \oplus M_P$.

We define the intersection multiplicity at P by

$$I(C_1 \cap C_2, P) := \dim_k\big(\mathcal{O}_P/(f_1, f_2)_P\big),$$

where $(f_1, f_2)_P$ denotes the ideal generated by f_1 and f_2 inside $\mathcal{O}_P$.

We now give an example illustrating that this definition matches geometric intuition. Consider the curves

$$\mathcal{C}_1 = \{y = 0\}, \qquad \mathcal{C}_2 = \{y = x^n\},$$

and compute their intersection multiplicity at $P = (0, 0)$. We take $f_1 = Y$ and $f_2 = Y - X^n$.

One difficulty is understanding quotients inside $\mathcal{O}_P$. To avoid this, we transport the computation back to R, modulo suitable powers of the maximal ideal.

Since it's clear that

$$\mathcal{M} = \{f \in R \mid f(P) = 0\} = (X, Y), \qquad \mathcal{M}_P = (X, Y)_P.$$

we have

$$\mathcal{M}^n = (X^n, X^{n-1}Y, \ldots, Y^n), \qquad \mathcal{M}_P^n = (X^n, X^{n-1}Y, \ldots, Y^n)_P.$$

and thus, the following decompositions

$$R = R_n \oplus \mathcal{M}^{n+1}, \qquad \mathcal{O}_P = R_n \oplus \mathcal{M}_P^{n+1}.$$

Indeed, for $h \in \mathcal{M}$ and $f \in R$, the element $f/(1-h)$ belongs to $\mathcal{O}_P$, and

$$\frac{f}{1-h} = f(1 + h + \cdots + h^n) + \frac{fh^{n+1}}{1-h} \in R + \mathcal{M}_P^{n+1}.$$

Moreover $R \cap \mathcal{M}_P^{n+1} = \mathcal{M}^{n+1}$, which yields

$$\mathcal{O}_P = R_n \oplus \mathcal{M}_P^{n+1}.$$

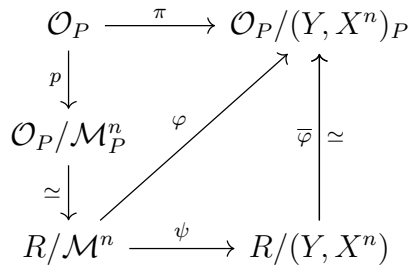
Thus, for every $n \geq 0$, we obtain a natural isomorphism

$$\boxed{\mathcal{O}_P/\mathcal{M}_P^{n+1} \simeq R/\mathcal{M}^{n+1}}.$$

Returning to our example, with $n \geq 1$, we have

$$(f_1, f_2)_P = (Y,\, Y - X^n)_P = (Y, X^n)_P, \qquad \mathcal{M}^n \subset (Y, X^n).$$

The computation relies on the following commutative diagram:



Here π is the natural projection modulo $(Y, X^n)_P$. Since $(Y, X^n)_P$ contains $\mathcal{M}_P^n$, the map π factors through $\mathcal{O}_P/\mathcal{M}_P^n$, and via the isomorphism above, through $R/\mathcal{M}^n$. Thus

$$\varphi : f \bmod \mathcal{M}^n \longmapsto f \bmod (Y, X^n)_P$$

is surjective, and its kernel is $(Y, X^n)/\mathcal{M}^n$. By the correspondence theorem,

$$\frac{R/\mathcal{M}^n}{(Y, X^n)/\mathcal{M}^n} \simeq R/(Y, X^n).$$

Hence ψ is the natural projection, and $\overline{\varphi}$ is an isomorphism. We conclude that

$$\boxed{\mathcal{O}_P/(Y, X^n)_P \simeq R/(Y, X^n)}.$$

Finally, the k-vector space $R/(Y, X^n)$ is isomorphic to $k[X]/(X^n)$, hence has dimension n. Therefore

$$\boxed{I(C_1 \cap C_2, (0,0)) = \dim_k\big(\mathcal{O}_P/(Y, X^n)_P\big) = n}.$$

We now record two basic properties of $I(C_1 \cap C_2, P)$ for $P \in k^2$:

- If $P \notin \mathcal{C}_1 \cap \mathcal{C}_2$, then one of the f_i is a unit in $\mathcal{O}_P$, so $(f_1, f_2)_P = \mathcal{O}_P$ and $I(\mathcal{C}_1 \cap \mathcal{C}_2, P) = 0$.
- If $P \in \mathcal{C}_1 \cap \mathcal{C}_2$, then $(f_1, f_2)_P \subset M_P$. Since $\mathcal{O}_P = k \oplus M_P$, we have

  $$\mathcal{O}_P/(f_1, f_2)_P \simeq k \oplus M_P/(f_1, f_2)_P,$$

  and therefore

  $$I(\mathcal{C}_1 \cap \mathcal{C}_2, P) = 1 + \dim_k\big(M_P/(f_1, f_2)_P\big) \geq 1,$$

  with equality if and only if $M_P = (f_1, f_2)_P$.

2.4 Step 4

We now have enough information on intersection multiplicity to prove

$$\sum_{P \in \mathcal{C}_1 \cap \mathcal{C}_2 \cap k^2} I(\mathcal{C}_1 \cap \mathcal{C}_2, P) \;=\; \dim\big(R/(f_1, f_2)\big).$$

We first show that $\dim(\mathcal{O}_P/(f_1, f_2)_P) \leq \dim(R/(f_1, f_2))$, which ensures that the intersection multiplicity is finite. Let $(g_i/h)_{i \leq r} \subset \mathcal{O}_P$ be a family linearly independent modulo $(f_1, f_2)_P$. Choose a common denominator h for all elements. If $(\lambda_i)_{i \leq r} \in k^r \setminus \{0\}$ satisfies

$$\sum_{i \leq r} \lambda_i g_i \in (f_1, f_2),$$

then

$$\sum_{i \leq r} \lambda_i g_i/h \in (f_1, f_2)_P,$$

which contradicts the independence modulo $(f_1, f_2)_P$. Thus $(g_i)_{i \leq r}$ is linearly independent modulo (f_1, f_2), and therefore

$$\dim(\mathcal{O}_P/(f_1, f_2)_P) \leq \dim(R/(f_1, f_2)).$$

Since $\mathcal{O}_P/(f_1, f_2)_P$ is finite-dimensional, it is generated by finitely many elements of the form g_i/h. Because $h^{-1} \in \mathcal{O}_P$, the numerators (g_i) also generate $\mathcal{O}_P/(f_1, f_2)_P$. Hence

$$\boxed{\mathcal{O}_P = R + (f_1, f_2)_P.}$$

We now show that for any integer $r \geq \dim(\mathcal{O}_P/(f_1, f_2)_P)$,

$$\mathcal{M}_P^r \subset (f_1, f_2)_P.$$

Let $g_1, \ldots, g_r \in \mathcal{M}_P$. Define a decreasing sequence of ideals

$$J_i := (g_1 \cdots g_i)\mathcal{O}_P + (f_1, f_2)_P \qquad (1 \leq i \leq r), \qquad J_{r+1} := (f_1, f_2)_P.$$

Since $\dim(\mathcal{O}_P/(f_1, f_2)_P) \leq r$, the chain stabilizes: there exists $i \in [\![1, r]\!]$ such that $J_i = J_{i+1}$.

If $i = r$, then $g_1 \cdots g_r \in (f_1, f_2)_P$ and we are done. Otherwise $i < r$, and by definition of $J_i = J_{i+1}$ there exist $\phi \in \mathcal{O}_P$ and $\psi \in (f_1, f_2)_P$ such that

$$g_1 \cdots g_i = g_1 \cdots g_{i+1}\, \phi + \psi.$$

Thus

$$g_1 \cdots g_i(1 - g_{i+1}\phi) = \psi \in (f_1, f_2)_P.$$

Since $g_{i+1} \in \mathcal{M}_P$, the element $(1 - g_{i+1}\phi)$ is a unit in $\mathcal{O}_P$, hence

$$g_1 \cdots g_r = \psi\, g_{i+1} \cdots g_r (1 - g_{i+1}\phi)^{-1} \in (f_1, f_2)_P.$$

Therefore $\mathcal{M}_P^r \subset (f_1, f_2)_P$.

We now prove

$$\dim(R/(f_1, f_2)) = \sum_{P \in \mathcal{C}_1 \cap \mathcal{C}_2 \cap k^2} I(\mathcal{C}_1 \cap \mathcal{C}_2, P).$$

Consider the natural map

$$\begin{array}{rccc} \psi : & R & \longrightarrow & \displaystyle\prod_{P \in \mathcal{C}_1 \cap \mathcal{C}_2 \cap k^2} \mathcal{O}_P/(f_1, f_2)_P \\ & f & \longmapsto & (f \bmod (f_1, f_2)_P)_P \end{array}.$$

Surjectivity. Since $\mathcal{C}_1 \cap \mathcal{C}_2 \cap k^2$ is finite (Step 1), we may choose polynomials $(g_P)_P$ such that

$$g_P(Q) = \delta_{P,Q}.$$

where δ is Kronecker's symbol. Fix $P \in \mathcal{C}_1 \cap \mathcal{C}_2 \cap k^2$ and $\phi \in \mathcal{O}_P$. Let $h = g_P$, so that $h^{-1} \in \mathcal{O}_P$ and $h \in \mathcal{M}_Q$ for $Q \neq P$. Because $\mathcal{O}_P = R + (f_1, f_2)_P$, there exists $f \in R$ such that

$$h^{-r}\phi \equiv f \mod (f_1, f_2)_P.$$

Set $g := h^r f \in R$. Then $g \equiv \phi \mod (f_1, f_2)_P$, and for $Q \neq P$ we have $h^r \in \mathcal{M}_Q^r$ and $\mathcal{M}_Q^r \subset (f_1, f_2)_Q$, hence

$$g \equiv 0 \mod (f_1, f_2)_Q.$$

Thus g maps to the element with ϕ in the P-component and 0 elsewhere. Since these elements form a basis of the product, ψ is surjective.

Kernel. Clearly $(f_1, f_2) \subset \ker(\psi)$. Let $f \in \ker(\psi)$ and define

$$L := \{g \in R \mid gf \in (f_1, f_2)\}.$$

Then L is an ideal of R containing (f_1, f_2). If $L \neq R$, the weak Nullstellensatz implies that there exists $P \in k^2$ such that

$$g(P) = 0 \quad \text{for all } g \in L.$$

But by construction of the g_P, there exists a polynomial not vanishing at P, contradicting $g_P \in L$. Thus $L = R$, so $1 \in L$, hence $f \in (f_1, f_2)$.

Therefore $\ker(\psi) = (f_1, f_2)$, and we conclude:

$$\boxed{\dim(R/(f_1, f_2)) = \sum_{P \in \mathcal{C}_1 \cap \mathcal{C}_2 \cap k^2} I(\mathcal{C}_1 \cap \mathcal{C}_2, P)}.$$

2.5 Step 5

In this section, we must work on the projective plane, and therefore we need to adapt our previous definitions so that they make sense at points at infinity.

We begin with points of k^2, and the definitions will then extend naturally to $\mathbb{P}^2$. Write $x = X/Z$ and $y = Y/Z$, and view

$$R = k[x, y] = k[X/Z,\, Y/Z] \subset k(X, Y, Z).$$

Then $K = k(x, y)$ identifies with the set of rational functions $\Phi = F/G$ where F and G are homogeneous of the same degree. Indeed, for $\phi \in K$,

$$\phi(x, y) = \frac{f(x, y)}{g(x, y)} = \frac{Z^n f(X/Z, Y/Z)}{Z^n g(X/Z, Y/Z)} = \frac{F(X, Y, Z)}{G(X, Y, Z)} = \Phi(X, Y, Z),$$

where $n = \max(\deg f, \deg g)$ and F, G are the homogenizations of f, g. Conversely, if $\Phi = F/G$ with F and G homogeneous of the same degree, then $\Phi(tX, tY, tZ) = \Phi(X, Y, Z)$ and

$$\Phi(X, Y, Z) = \Phi(x, y, 1) \in K.$$

We define

$$\mathcal{O}_P := \{\Phi = F/G \in K \mid G(P) \neq 0\}, \qquad \mathcal{M}_P := \{\Phi \in \mathcal{O}_P \mid \Phi(P) = 0\}.$$

Here Φ is always taken in K, so it is homogeneous of degree 0. If $P \in k^2$, these definitions coincide with the affine ones, since $k^2 \simeq \mathbb{P}^2 \cap \{Z \neq 0\}$.

Let $\mathcal{C}_i = \{P \in \mathbb{P}^2 \mid F_i(P) = 0\}$ for $i = 1, 2$, where $F_i \in k[X, Y, Z]$ is homogeneous of degree n_i, and let $f_i(X, Y) = F_i(X, Y, 1)$ be its affine part. We define

$$(F_1, F_2)_P := \{\, F \in \mathcal{O}_P \mid F = H_1F_1 + H_2F_2 \text{ for some } H_1, H_2 \in \mathcal{O}_P \,\}.$$

We do not simply take the ideal generated by F_1 and F_2 in K, because we must keep rational functions homogeneous of degree 0 in order for them to be well defined on $\mathbb{P}^2$. Equivalently,

$$(F_1, F_2)_P = \mathcal{O}_P \cap (\mathcal{O}_P F_1 + \mathcal{O}_P F_2).$$

Finally, if $P \in \mathbb{P}^2 \cap k^2$, then

$$(f_1, f_2)_P = (F_1, F_2)_P,$$

because each f_i is a polynomial in X/Z and Y/Z, and both Z and Z^{-1} belong to $\mathcal{O}_P$.

Since $\mathcal{O}_P$ naturally embeds into K, and rational functions in K are invariant under linear changes of projective coordinates, the definitions of $\mathcal{O}_P$ and $(f_1, f_2)_P$ are also invariant. Therefore the intersection multiplicity $I(\mathcal{C}_1 \cap \mathcal{C}_2, P)$ is independent of the chosen projective coordinate system.

In Step 1 we proved that $\mathcal{C}_1 \cap \mathcal{C}_2 \cap k^2$ is finite. This remains true after changing the line at infinity. We now prove that $\mathcal{C}_1 \cap \mathcal{C}_2$ is finite. Let L and L' be two distinct lines in $\mathbb{P}^2$. Each of them misses only finitely many points of $\mathcal{C}_1 \cap \mathcal{C}_2$, and since L and L' intersect in exactly one point, the set $\mathcal{C}_1 \cap \mathcal{C}_2$ must be finite.

Because k is algebraically closed, it is infinite. Hence there exists a line in $\mathbb{P}^2$ that does not meet $\mathcal{C}_1 \cap \mathcal{C}_2$; otherwise the set of directions (i.e. $\mathbb{P}^1$) would be finite. Let $\mathcal{L}$ be such a line, and choose coordinates so that $\mathcal{L}$ becomes the line at infinity. By construction, $\mathcal{C}_1$ and $\mathcal{C}_2$ do not meet at infinity.

Using Step 2, we obtain

$$\boxed{\dim(R/(f_1, f_2)) = n_1 n_2}.$$

By Steps 4 and 5, the definition of intersection multiplicity is valid in $\mathbb{P}^2$, and therefore

$$\dim(R/(f_1, f_2)) = \sum_{P \in \mathcal{C}_1 \cap \mathcal{C}_2} I(\mathcal{C}_1 \cap \mathcal{C}_2, P).$$

Thus we conclude that

$$\boxed{\sum_{P \in \mathcal{C}_1 \cap \mathcal{C}_2} I(\mathcal{C}_1 \cap \mathcal{C}_2, P) = n_1 n_2}.$$

This completes the proof of Bézout's theorem for plane projective curves.